

PERSHORE, United Kingdom

WMO: 035290

Lat: 52.1483N

Lon: 2.0411W

Elev: 102

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 23.3	(c) 26.3	(d) 21.5	(e) 16.2	(f) 27.9	(g) 23.9	(h) 18.2	(i) 30.1	(j) 27.5	(k) 48.1	(l) 25.2	(m) 48.7	(n) 2.1	(o) 340	(p) 0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.7	(c) 80.0	(d) 65.0	(e) 76.3	(f) 63.4	(g) 73.3	(h) 62.1	(i) 66.7	(j) 76.1	(k) 65.0	(l) 73.3	(m) 63.5	(n) 70.7	(o) 7.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.7	69.8	61.8	82.8	68.1	60.4	78.7	66.7	31.3	76.2	30.0	73.7	28.9	70.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.7	(b) 20.0	(c) 17.7	(e) 16.9	(f) 86.5	(g) 5.0	(h) 4.1	(i) 13.3	(j) 89.5	(k) 10.3	(l) 91.9	(m) 7.5	(n) 94.2	(o) 3.9	(p) 97.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			17.4	71.0	5.4	3.4	13.5	73.4	10.3	75.4	7.3	77.3	3.4	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	40.6	41.4	44.1	48.2	53.6	58.9	62.8	62.1	57.8	51.8	45.5	41.0		
	DBStd	9.70	6.40	6.00	5.55	5.33	5.29	4.68	4.35	4.25	4.39	5.53	6.39	7.24		
	HDD50	1345	295	246	194	94	24	1	0	0	2	45	159	285		
	HDD65	5282	758	662	647	504	355	191	97	110	219	410	586	743		
	CDD50	1598	3	4	11	40	134	268	396	376	236	100	23	7		
	CDD65	61	0	0	0	0	1	8	28	21	3	0	0	0		
	CDH74	583	0	0	0	2	21	102	290	140	25	3	0	0		
	CDH80	102	0	0	0	0	0	19	58	24	1	0	0	0		
(14) Wind	WSAvg	7.8	9.2	9.2	8.7	7.9	7.4	7.0	6.7	6.9	6.9	7.8	7.8	8.4		
(15) Precipitation	PrecAvg	24.9	2.3	1.5	1.8	1.9	2.1	2.3	1.8	2.4	2.4	2.0	2.1	2.4		
	PrecMax	31.2	4.1	4.0	4.3	5.7	6.0	6.5	4.4	5.0	6.9	5.0	5.2	5.0		
	PrecMin	17.1	0.4	0.2	0.2	0.2	0.4	0.2	0.2	0.3	0.3	0.2	0.9	0.5		
	PrecStd	3.3	1.0	1.0	0.9	1.1	1.3	1.6	1.1	1.3	1.5	1.1	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	57.6	62.7	71.2	76.8	82.8	85.8	83.4	77.5	68.4	60.6	57.4		
		MCWB	52.1	52.2	52.2	57.9	62.4	65.7	66.0	66.5	63.5	60.9	56.8	53.8		
	2%	DB	53.7	54.1	58.4	65.0	72.1	76.5	81.0	77.6	72.5	64.4	58.4	55.4		
		MCWB	50.4	50.1	51.0	54.5	60.6	63.4	65.4	64.5	61.8	58.2	55.4	52.9		
	5%	DB	51.8	52.2	55.7	61.2	67.8	72.5	77.0	74.0	69.3	62.2	56.7	53.6		
		MCWB	49.0	48.7	49.7	52.5	58.1	61.6	63.7	62.9	60.1	57.1	53.7	51.0		
	10%	DB	49.9	50.3	53.4	58.0	63.9	69.1	73.5	71.0	66.6	60.2	54.7	51.6		
		MCWB	47.2	47.0	48.6	50.8	55.8	59.9	62.1	61.2	58.9	56.0	51.8	49.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.0	54.4	58.9	64.7	69.3	70.5	68.2	66.1	62.4	57.9	54.7		
		MCDB	55.0	56.1	59.1	68.4	73.9	79.4	80.9	78.6	74.5	67.3	59.8	56.4		
	2%	WB	51.2	50.9	52.4	55.8	61.6	64.8	66.7	66.2	63.6	60.0	55.9	53.1		
		MCDB	53.2	53.5	56.7	62.8	69.8	73.2	77.0	74.6	69.6	63.2	58.0	55.0		
	5%	WB	49.3	49.2	50.8	53.7	59.2	62.7	65.0	64.6	61.8	58.3	54.0	51.4		
		MCDB	51.5	51.8	54.7	59.8	65.8	70.1	74.3	71.5	66.9	61.4	56.2	53.4		
	10%	WB	47.5	47.4	49.0	51.6	57.0	60.8	63.3	63.0	60.2	56.5	52.2	49.2		
		MCDB	49.7	50.1	52.9	57.1	62.7	67.5	71.2	68.8	65.0	59.5	54.3	51.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.6	14.5	16.8	17.1	17.3	17.7	16.9	16.4	13.7	11.4	10.3		
		MCDBR	10.9	12.4	17.4	22.1	23.7	23.6	24.6	21.2	21.0	14.2	10.7	10.4		
		MCWBR	9.9	9.8	11.6	13.3	13.7	12.5	11.6	10.5	11.9	9.7	8.9	9.5		
	5% WB	MCDBR	10.8	12.3	15.5	19.8	20.7	20.1	21.2	17.8	17.3	12.7	10.8	10.7		
		MCWBR	10.2	10.3	11.0	12.6	12.7	11.9	11.7	9.9	11.0	9.7	9.6	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.339	0.372	0.397	0.402	0.398	0.395	0.382	0.368	0.351	0.326	0.313		
	taud		2.358	2.382	2.287	2.229	2.250	2.292	2.303	2.363	2.402	2.427	2.408	2.334		
	Ebn at Noon		212	246	258	265	269	272	270	267	258	237	211	193		
	Edh at Noon		20	25	33	40	41	40	38	34	30	24	18	17		
(44) All-Sky Solar Radiation	RadAvg		241	454	803	1233	1519	1638	1546	1274	961	550	290	190		
	RadStd		21	54	100	141	91	171	163	100	79	49	31	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (97 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page